A-A-53777
10 December 1988

COMMERCIAL ITEM DESCRIPTION

AUDIOMETER, Manually Operated, Portable,
110/220 Volt, 50/60 Hz, AC

The General Services Administration has authorized the use of this commercial item description in preference to Military Specification MIL-A-36494, which is cancelled.

This commercial item description covers a single channel, fixed frequency oscillator, portable, manually operated, audiometer capable of performing basic threshold evaluations with calibrated masking noise and bone conduction.

Salient characteristics.

General. The audiometer shall be capable of performing the following pure tone hearing tests:

Pure tone air conduction.

Pure tone air conduction with masking.

Pure tone bone conduction.

Pure tone bone conduction with masking

The audiometer shall comply with American National Standards Institute (ANSI) Standard S3.6. In the event of confliction, the requirements specified herein shall take precedence.

Exceptions. The following portions of ANSI Standard S3.6 do not apply:

4.2 Test Tones for Limited Range and Narrow Range Audiometers

4.3 Variation in Method of Presentation of Test Tones

5. Requirements for Speech Audiometers

AMSC/NA FSC 6515

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Test Tones. The audiometer shall generate test tones of at least the following fixed frequencies:

Air Conduction (Hz)	Bone Conduction (Hz)
250	250
500	500
1000	1000
1500	1500
2000	2000
3000	3000
4000	4000
5000	
6000	
8000	

Accessories. The unit shall be provided with the following accessories:

Qty	Description
1 Pr	Air Receivers
1 Set	Head Band and Cord
1 Set	Pator Cord for (See Note 2) Audiometer Booth
1 Ea	Bone Oscillator with Headband and Cord
6 Ea	Spare Fuses (if applicable)
1 Pr	Additional Air Receiver Cord

Instructions. The unit shall be provided with 2 sets of Instruction and Service manuals. A list of high mortality repair parts and wiring diagrams shall be included.

Electrical. The audiometer shall be capable of operating from the following line sources: 110 volts (plus or minus 10 percent) and 220 volt (plus or minus 10 percent), multiple frequency, 50 and 60 Hz (plus or minus 2 Hz). The unit shall be supplied with a UL-approved three-wire ground line cord and a 110 volt green dot hospital grade male plug.

Carrying Case. The audiometer shall be contained in a rugged shock resisting carrying case with handle. The case shall offer a protective cover for the control panel and be capable of housing all accessory components including Instruction data.

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Calibration. Using the methods described in ANSI S3.6, the audiometer shall be calibrated to conform with specified requirements. A certificate of calibration shall be provided with the unit.

Workmanship. The audiometer shall be free from defects which detract from its appearance or impair its serviceability.

Unit. Each (EA). One audiometer, as specified, constitutes one unit.

Quality Assurance Provisions.

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspection set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

Inspection. Inspection, as used herein, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

First article inspection. First article inspection shall be performed on the audiometer when a first article sample is required. This inspection shall include the examination and all tests. All tests are to be performed using each specified input power voltage and frequency.

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Quality conformance inspection. Quality conformance inspection shall include the examination and tests specified herein.

Quality Conformance Sampling.

Sampling for inspection. Sampling for examination and testing shall be conducted in accordance with MIL-STD-105, with an inspection level of S-2 and an AQL of 1.5 (percent defective). The unit of product for sampling purposes shall be one audiometer.

Examination and Testing.

Examination. The audiometer shall be examined to determine compliance with the requirements of this specification in accordance with the defects listed below.

Defects

Any component part including accessories missing, broken or otherwise unusable.

Exposed edge not free of burrs or sharpness.

Identification marking not complete, permanent, and legible.

Finish not free of chips, cracks, scratches and other appearance defects.

Any component not secured in place.

Testing. Testing shall be conducted to determine compliance with specified requirement. First article testing requires all tests listed herein. Quality Conformance testing requires the first three tests at all the specified voltages and frequencies. Testing shall be performed after a 30 minute warm-up period. Testing shall be conducted in accordance with the methods outlined in ANSI S3.6.

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Tests

Air Conduction Tests.

1. Headband provided is adequate to hold the earphones against the ears to provide a satisfactory seal.

2. Each of the frequencies generated are within three percent of the indicated frequency.

3. Sound pressure level of any harmonic is at least 30 dB below the sound pressure level of the fundamental.

4. Dial readings of hearing threshold levels have a minimum range of from 0 dB to the values given in Table II (ANSI S3.6), by intervals of 5 dB or less, with the 0 dB setting at each frequency corresponding to the reference threshold sound pressure levels stated under defects.

5. The sound pressure produced by an earphone as referred to the standard reference level does not differ from the indicated value of sound pressure level at any reading of the hearing threshold level dial by more than 3 dB at indicated frequencies of 250 to 3000 Hz inclusive, by more than 4 dB at 4000 Hz, nor by more than 5 dB at frequencies above or below this range.

6. Any sound from the test earphone, other than the desired test signal, is not of a magnitude as to affect threshold judgement.

7. Any signal from the earphone not being used for test purposes has a sound pressure level at least 10 dB below the hearing threshold reference level, except that it need not be more than 70 dB below the signal from the "on" earphone.

8. Any sound due to the operation of the audiometer controls during the actual listening test is inaudible at each setting of the hearing threshold level dial up to and including 50 dB.

9. Tone switch operation does not produce audible transients of extraneous frequencies.

10. Rise and fall times are as specified in ANSI S3.6 paragraph 4.5.

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11. Steady output voltage level in the tone switch "off" position is at least 50 dB below its steady voltage in the "on" position, or at least 10 dB below that corresponding to the zero Hearing Threshold Level, whichever is higher.

Bone Conduction Tests.

At any test frequency of 2000 Hz or lower and as far as possible at higher frequencies, the bone vibrator does not radiate sound to such an extent that the sound reaching the ear by air conduction impairs the validity of the bone conduction measurement.

The headband provided holds the bone vibrator in position.

Each of the frequencies generated are within three percent of the indicated frequency.

The masking sound is at least sufficient to mask the tones produced when the dial reading is 40 dB hearing threshold level at 250 Hz, 50 dB at 500 Hz, and 60 dB at 1000 Hz and higher test frequencies.

Contractor certification. The contractor shall certify that the product offered meets the salient characteristics of this description and conforms to the producers' own drawings, specification, standard, and quality assurance practices. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Metric Products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, providing they fall within the tolerances specified using conversion tables contained in the latest revision of Federal Standard No. 376, and all other requirements of this document are met.

If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch pound units, a request should be made to the contracting officer to determine if the product is acceptable.

The contracting officer has the option of accepting or rejecting the product.

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Regulatory requirements.

Federal Food, Drug and Cosmetic Act. If the product covered by this document has been determined by the U.S. Food and Drug Administration to be under its jurisdiction, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug and Cosmetic Act, as amended, and regulations promulgated thereunder. In addition, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal, State, and local statutes, ordinances, and regulations.

Recovered materials. The offeror/contractor is encouraged to use recovered material in accordance with Public Law 94-580, as amended, and as implemented by Federal Acquisition Regulation Subpart 23.4 to the maximum extent practical.

Preservation, packaging, packing, labeling, and marking. Unless otherwise specified, preservation, packaging, and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling, etc., involving shipment from the supply source to the receiving activity, plus reshipment from receiving activity, and shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities and labeling and marking shall be as specified in the contract and/or order.

NOTE 1: The following National Stock Number is covered by this document:

NSN	ITEM IDENTIFICATION
6515-00-926-1520	AUDIOMETER, MANUALLY OPERATED, PORTABLE, 110/220 Volt, 50/60 Hz

NOTE 2: The audiometer shall be compatible with Patch Cord and Audiometer Booth (NSN. 6515-00-065-9600). The audiometer booth has a jack panel which includes: six 3-wire phone jacks, one cinch jones 303 connector and one cinch jones 304 connector.

Ordering data. Intermediate/exterior package quantities and labeling and marking must be specified in the contract and/or order.

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MILITARY INTERESTS:

Custodians:

Army - MD
Navy - MS
Air Force - 03

Preparing Activity:

DoD-MB

Agent:

DLA-DH

CIVIL AGENCY COORDINATING ACTIVITIES:

VA-OSS
PHS
FDA-MPQAS

Project No. 6515-4288

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1 DOCUMENT NUMBER	2 DOCUMENT TITLE
A-A-53777	AUDIOMETER, Manually Operated, Portable, 110/220 Volt,

3a NAME OF SUBMITTING ORGANIZATION 50760 AC	4 TYPE OF ORGANIZATION (*Mark one*)
	☐ VENDOR
	☐ USER
b ADDRESS (*Street, City, State, ZIP Code*)	☐ MANUFACTURER
	☐ OTHER (*Specify*)

5 PROBLEM AREAS

a Paragraph Number and Wording

b Recommended Wording

c Reason/Rationale for Recommendation

6 REMARKS

7a NAME OF SUBMITTER (*Last, First, MI*) – Optional	b WORK TELEPHONE NUMBER (*Include Area Code*) – Optional
c MAILING ADDRESS (*Street, City, State, ZIP Code*) – Optional	8 DATE OF SUBMISSION (*YYMMDD*)

DD FORM 1426 PREVIOUS EDITION IS OBSOLETE

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NOTE. This form may not be used to request copies of documents nor to request waivers, deviations or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

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